

ColdTrack AI: Evaluation Artifact (Iteration 2)

AIC COMPFEST 18 Final · Smart Logistics · results tags: iter1 → iter2 · follows Evaluation_Artifact_Iteration.pdf

What is this document?

Iteration 1 replaced the qualifying-round prototype with a physics engine plus a learned model, and fixed its worst failure: urgent cases were no longer shown as green. It left others open, listed as N1 to N7 in the iteration 1 artifact. Iteration 2 changes four rules in the physics engine and the sensor check to address N1, N2 and N6. No model was retrained, so the forecast and the event names (N3, N4, N5) are unchanged.

We measured iteration 2 twice. First on the same 106 test trips as the baseline and iteration 1 (16,336 snapshots from the old generator, dataset v4), so all three versions can be compared. Then on 225 held-out trips from the new physics simulator (dataset v3, trips the models never trained on). The second test exists because we chose the rule settings while looking at dataset v4; if a change only fitted v4, it would show up as a loss on dataset v3.

Summary

Iteration 2 passes 7 of 19 checks, the same count as iteration 1 (7), but the numbers behind them moved:

What we measure (dataset v4, server)	Baseline	Iteration 1	Iteration 2	Goal
Urgent cases (breach within 30 min) shown as green	32%	0%	2%	5% or less
Urgent cases shown as critical	25%	0%	16%	no goal set
Urgent cases with a countdown	62%	0%	23%	80% or more
Healthy snapshots with a warning or critical status	6%	33%	19%	10% or less
Sensor frozen during a real breach: flagged	10%	0%	37%	50% or more
Urgent cases treated as safe with 0.3 °C sensor noise	82%	50%	3%	50% or less
Checks passed	8 of 23	7 of 19	7 of 19	lists differ from baseline

On the new simulator's held-out trips nothing got worse for healthy trucks or sensor faults, and the critical status arrives earlier (median 44 → 54 minutes before the breach). The price is more alarms on door openings that never lead to a breach: critical status on 48% → 63% of those trips (section 3).

Checks that changed result: none.

1. What changed

Change	Finding it targets	Rule now
Drift countdown	N1: urgent cases got a warning but never a countdown or a critical status	When the smoothed sensor reading has risen at least 0.3 °C per hour over 20 minutes and the door has been shut for 30 minutes, the engine also counts the minutes until the reading itself would cross the limit, and uses the shorter of this and the physics countdown.
Margin scaled to the cargo range	N2: healthy trucks warned in 33% of snapshots	The “near the limit” warning starts at 35% of the cargo range width, capped at 1 °C (0.7 °C for 2 to 4 °C cargo; 1 °C for vaccine, fish and frozen meat as before). Section 3 shows why this value.
Sensor reading as upper bound	found while testing the margin change	The margin rule looks at the higher of the estimated cargo temperature and the smoothed sensor reading. Without this, the extreme-heat demo showed green with the sensor at 4.1 °C against a 4.0 °C limit, because the estimate for 1,000 kg of cargo lagged at 3.05 °C.
Stuck sensor after a rise	N6: a sensor frozen 30 minutes before a breach was never flagged	A flat reading is suspect once the temperature trend of the 30 minutes before it would have moved it by 0.3 °C.

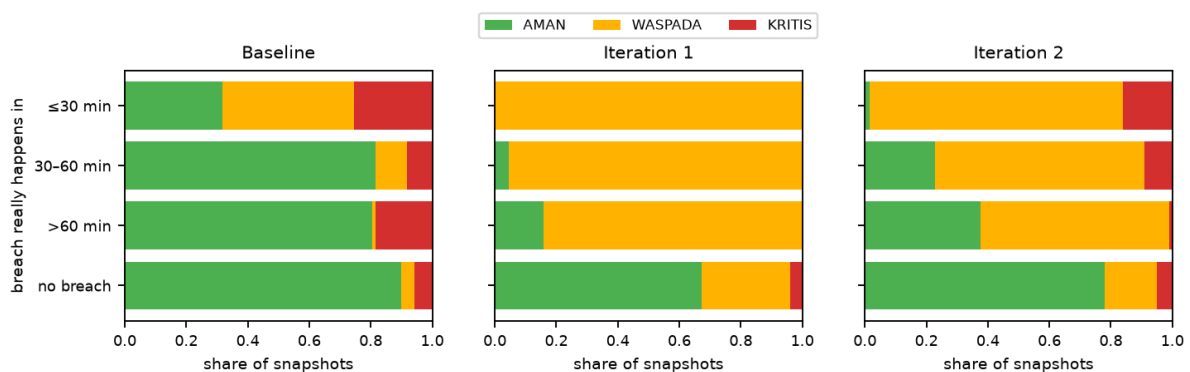
The countdown and the margin rule now treat the sensor reading as an upper bound on the cargo temperature. That is the safe choice while one question stays open: the engine was designed for a sensor in the cabin air, but the API documents temp_c as the cargo temperature. If the sensor sits in the cargo, the readings are the cargo temperature and these rules are exact. If it sits in the air, they warn earlier than needed, which is the cost described in section 3.

Demo scenarios (what the frontend shows)

Scenario	Expected	Iteration 2 status	Countdown	Diagnosis
Skenario 1: Perjalanan Normal (Kondisi Aman)	AMAN	AMAN	none	normal_sehat
Skenario 2: Pintu Kargo Terbuka Terlalu Lama	KRITIS	KRITIS	21 min	pintu_terbuka
Skenario 3: Kompresor Melemah Bertahap	KRITIS	KRITIS	18 min	suhu_muatan_mendekati_batas
Skenario 4: Sensor Macet (Stuck-at)	WASPADA	WASPADA	none	masalah_sensor
Skenario 5: Kejut Suhu Ambien saat Macet	WASPADA	WASPADA	none	suhu_muatan_mendekati_batas

The compressor scenario is critical again with an 18-minute countdown, its original design; iteration 1 had lowered it to warning because the physics countdown could not see a slow drift.

2. Results on the baseline test trips (dataset v4)



Each bar groups snapshots by how soon the cargo really leaves its safe range; the top bar should be red. Iteration 1 turned the baseline's green into amber; iteration 2 turns part of it red and brings the healthy bar back toward green.

Countdown

The countdown now appears in 22% of urgent snapshots (iteration 1: 0%), and when shown it is off by 9.5 minutes on average. In the rest the physics countdown was still above the 60-minute display limit (78%) or the sensor was flagged as broken (0%). Where the reading stays flat or falls until shortly before the breach, as in the cases below, there is no trend to project.

Urgent cases still shown as green

2% on the server sample, 4% over all urgent snapshots. We looked at the ones the narrower margin exposed: all came from 4 trips, the reading sat 0.54 to 0.84 °C below the limit with no upward trend, and it crossed within 9 to 30 minutes. Their temperature swing was smaller than that of healthy snapshots (median 0.10 against 0.12 °C), so nothing we measure tells them apart from a healthy truck at that moment. Only the margin width catches them (section 3).

Stuck sensor

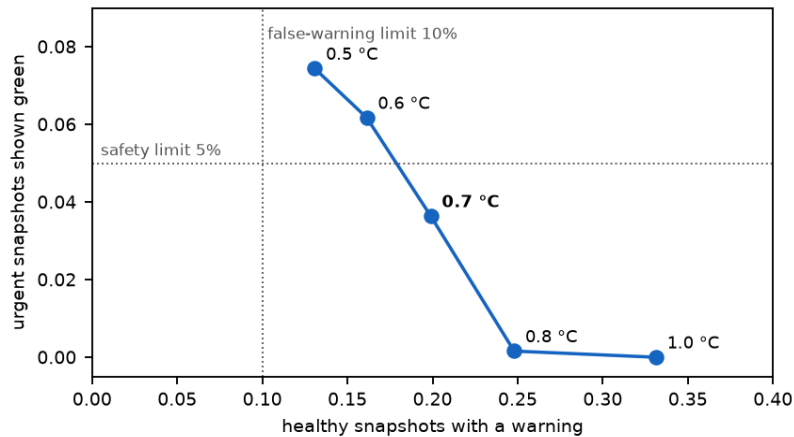
Flagged in 37% of the frozen-sensor cases (iteration 1: 0%), still below the 50% goal. Status is green in 10%. The new rule fires only when the reading was rising before it froze, so freezes after a flat stretch are still missed. Faulty sensors in general are named in 67% of cases (iteration 1: 62%), with no false sensor alarms.

Unchanged by design

Forecast error 1.41 °C, door named 0%, outside heat named 15%: these come from the learned model, which was not retrained. Server response time 120 ms at the 95th percentile, no errors in 1,634 requests.

3. Trade-offs

How wide should the “near the limit” margin be?



Margin for 2 to 4 °C cargo	Urgent shown green	Healthy warned
0.5 °C	7%	13%
0.6 °C	6%	16%
0.7 °C (chosen)	4%	20%
0.8 °C	0%	25%
1.0 °C	0%	33%

No width meets both goals on dataset v4: urgent and healthy snapshots sit at overlapping distances from the limit. 0.7 °C is the narrowest width that keeps urgent green at 5% or less (4%); 0.5 °C would cut healthy warnings to 13% but show 7% of urgent cases as green. The team chose safety. All values in the table are over all urgent and healthy snapshots, not the server sample.

Results on the new simulator's held-out trips (dataset v3)

Per trip	Iteration 1	Iteration 2	Trips
Breaching trips warned before the breach	100%	100%	76
Breaching trips critical before the breach	92%	95%	76
Median lead of the critical status (minutes)	44	54	
Healthy trips with any warning	0%	0%	58
Door trips without a breach: any warning	81%	96%	27
Door trips without a breach: critical	48%	63%	27
Sensor-fault trips flagged	99%	99%	75
Other trips wrongly flagged as sensor fault	0%	0%	

The door trips are where iteration 2 costs something. In the new simulator the sensor measures cabin air, which jumps when the door opens and can drift toward the limit while a heavy load keeps the cargo itself safe. Treating the reading as an upper bound turns those moments into warnings. When we checked the door trips that reached critical without a breach, most were raised by the door and physics rules that iteration 1 already had, and a few by the new drift countdown on loads of 1.8 to 3.3 tonnes. Knowing where the sensor sits would settle whether these are false alarms.

4. Still open

Finding	Status after iteration 2	Next step and how we will judge it
N1 countdown	shown in 22% of urgent cases (goal 80%)	Use the learned countdown (XGBoost) when neither physics nor drift can project; judge on both datasets.
N2 false warnings	19% of healthy snapshots (goal 10%)	Needs a signal that separates slow approaches from healthy running; the margin alone cannot (section 3).
N3 forecast	1.41 °C error, unchanged	Retrain to predict the change, not the level.
N4 door / outside heat	unchanged	Add wall and rack heat storage and a sunlight term to the simulator, retrain.
N5 noise	event names still collapse at 0.1 °C noise; status now holds (urgent green 3% at 0.3 °C)	Train with 0.05 to 0.5 °C noise.
N6 stuck sensor	flagged 37% (goal 50%), green 10%	A flat reading while the estimated cargo should be warming, without needing a prior trend.
N7 test validity	second test set added (dataset v3 held-out trips)	Keep both; report every change on both.
Sensor position	open question	Confirm whether the IoT probe sits in the cargo or in the cabin air; it decides the door-trip trade-off.

Appendix: acceptance checks (iteration 2)

Result	Check
PASS	test_recall_per_event_at_least_0_5[sensor]
PASS	test_imminent_breach_not_shown_as_safe
PASS	test_sensor_noise_0_3C_does_not_hide_imminent_breach
PASS	test_timezone_shift_does_not_change_event_recall_much
PASS	test_imminent_breach_never_green_more_than_5pct
PASS	test_api_no_http_errors_and_latency_p95_under_300ms
PASS	test_api_matches_batch_scoring
FAIL	test_forecast_mae_t30_under_0_8C
FAIL	test_forecast_worst_case_error_bounded
FAIL	test_recall_per_event_at_least_0_5[door]
FAIL	test_recall_per_event_at_least_0_5[shock]
FAIL	test_healthy_windows_not_labelled_as_fault
FAIL	test_imminent_breach_ttb_shown_at_least_80pct
FAIL	test_test_split_covers_every_cargo_profile_for_imminent_breach
FAIL	test_sensor_noise_0_3C_event_recall_drop_at_most_0_05
FAIL	test_stuck_sensor_during_breach_is_flagged_not_safe
FAIL	test_imminent_breach_ttb_shown_by_api_at_least_80pct
FAIL	test_healthy_windows_not_over_alerted
FAIL	test_every_fault_mode_has_at_least_5_test_trips

Reproduce: `python -m eval.run_eval --tag iter2`; `python -m eval.run_eval --tag iter2 --margin-sweep`; `python -m eval.sim_check --tag iter2` (needs `ml/dataset_v3.parquet`, not in git); `EVAL_TAG=iter2 pytest eval/test_baseline.py`; `python -m eval.make_artifact --iteration 2`. Iteration 1 simulator numbers were produced with the engine at commit `ff9aacd`.

Code commit: `62e776e`. Datasets: `data/processed/v4_seed1000_700trips.parquet` (split=test); `ml/dataset_v3.parquet` (test split of `prep_windows.split_trips`, seed 0). Cargo mass on v4: profile default (1,000 kg).